QPS TRIAGE · ABACUS · CODEX · FEDERATED DMAIC

NOT LIFTED · GATE NOT RESTORED

# Global Code Purple

A recursively self-testing execution protocol across repositories, agents, CI, PR admission and evidence. It builds on QPS TRIAGE, ABACUS, CODEX, KEB and DOW and adds no second authority.

| | |
|---|---|
| STATUS AT | GitHub readback 2026-10-18 13:17 UTC · not lifted: 2 of 5 lift criteria met |
| SEED | CODEX main · federation/mesh/CODE_PURPLE_EXECUTION_SEED_20261012.yaml |
| OWNER | GBO · Process Engineer, QPLANT System Owner, SCK CEN |

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EXECUTIVE SUMMARY

# ABACUS has merged nothing since 6 October; one PR unblocks it

### WHAT IS BLOCKED

All 20 open ABACUS PRs. The **Main Hard Gate** ruleset requires one check, `PR Feedback Gate`, which waits for an exact-head Codex code and security review. The Codex allowance is exhausted, so the gate never turns green.

### THE FIX IN FLIGHT

**ABACUS #1858** lets the owner attest the exact head when Codex cannot run. It cannot approve itself: the gate runs main's code. It needs a one-time owner **break-glass** on the ruleset to land.

### ALREADY MERGED

**CODEX #984** (this protocol's seed) merged at 12:16 with 4 red checks. CODEX main has no required-check ruleset, so nothing stopped it. CODEX main is green again on its current head.

**Decisions needed from GBO:** (1) accept the 4 inherited reds under break-glass (they fail nightly on main) · (2) add the one-time bypass on ruleset 24494381. Done: threads resolved, #1878/#1879 opened, policy-and-inventory fixed (e7616e62).

READBACK

# Tracked heads at 17:55

| ITEM | SUBJECT | HEAD | STATE | PASS / FAIL / NEUTRAL | RED CHECKS |
|---|---|---|---|---|---|
| **ABACUS #1858** | Owner-attested review fallback | e7616e62 | open | 87 / 2 / 1 | Code Quality, Security Scanning (Coverage, Full Red Proof still running) · policy-and-inventory now green |
| **ABACUS #1859** | PR-level new-debt ratchet | 6fac58cc | open | 61 / 4 / 1 | Code Quality, Full Red Proof, Security Scanning, validate-scripts |
| **ABACUS #1857** | Fix syntax error in diagnostic_kpi_probe.py | e7b45912 | open | 60 / 4 / 1 | Code Coverage, Code Quality, Full Red Proof, Security Scanning |
| **ABACUS #1856** | Lossless branch hygiene burn-down | f11d8044 | open | 91 / 3 / 1 | Code Coverage, Code Quality, Full Red Proof |
| **CODEX #984** | Code Purple execution seed | 90e3396a | merged 12:16 | 55 / 4 / 0 | ruff, governance-gate ×2, validate (at merge) |
| **CODEX main** | after #984 merge | 768c9e35 | main | 38 / 2 / 0 | validate (fixed on main at f9ae4676), osv-scan (fails on every main push since before #984) |
| **CODEX main** | current head | cce9ce20 | main | — | MissionControl CI green; OSV-Scanner red (pre-existing) |

Counts exclude skipped and cancelled runs. ABACUS main's last merge is #1838 (2026-10-06).

TAXONOMY

# Four kinds of reference, one model

## STRUCTURE · WHAT THE PROTOCOL DOES

**1–8 Stages**
The pulse written as a procedure. Each stage groups nodes and lists entry, actions and exit.

contain ↓

**N01–N15 Nodes**
One step of a pulse: who owns it, which artefact, what evidence it writes. Drawn on the pulse graph.

are joined by ↓

**E01–E19 Edges**
Typed connections: sequence, pass, red, hold, evidence, recursion.

## LIFECYCLE · WHAT MOVES A WORK ITEM

**12 States**
Where a PR head or red finding is: test pending, set-up, run, debug, fix, integrate, admission, follow-up, hold, merged, federate.

are changed by ↓

**T01–T16 Triggers**
One event, one guard, one action. Each trigger names the nodes it acts at and whether it is wired today.

act at → Nodes

## EVIDENCE · WHAT PROVES IT

**per node Telemetry**
Preconfig (stage, test, debug), run attempts with pass / fail / neutral / pending, errors, logs.

is kept as ↓

**sha Receipts & census**
Exact-commit receipts per run; one census row per iteration N.

is learned as ↓

**REX Lessons**
Recurring causes and reusable fixes, read back at the start of the next pulse.

**Worked example.** Trigger **T11** moved #1858 from *Admission* to *HOLD*. It acts at nodes **N10** (gate) and **N11** (hold), which belong to stages **6** and **7**. On the pulse graph that move is edge **E12**, and the hold record is evidence for REX.

Phase letters **D M A I C** (Define, Measure, Analyse, Improve, Control) tag every node and stage.

LEGEND

# Colour, line and status vocabulary

## NODE ROLES (FILL)

**Authority**
Resolves or owns state: controller, allocation, output

**Process**
Checks and verification

**Gate**
A binary decision

**Repair**
Isolated change and retest

**Admitted path**
Merge and post-merge proof

**Hold**
Blocked, logged, retried

**Evidence store**
REX and KEB

## EDGE TYPES (LINE)

**Sequence**
Normal next step

**Pass**
Admissible successor

**Material red**
Goes to repair

**Hold**
Deferred or blocked

**Evidence**
Receipts and lessons

**.self recursion**
Output N becomes input N+1

## MARKERS

**Receipt**
Step writes an exact-commit receipt

**Run bar**
pass · fail · pending share of attempts

**Tracked PR**
Where a live PR is now

## IMPLEMENTATION STATUS

Seeded / committed

In progress

Partial

On hold

Not implemented

Not reached

## TRIGGER WIRING

Wired — Fires automatically today

Partly wired — Detection automatic, response manual

Manual — A person does it

Proposed — Not built yet

GLOSSARY

# Terms used in this deck

| | |
|---|---|
| DMAIC | Define, Measure, Analyse, Improve, Control: the phases each node belongs to. |
| .self | The recursive interface: validated output N becomes input N+1 through governed reconciliation (⊕). |
| Material red | A completed failing check that blocks admission, as opposed to neutral or pending. |
| Break-glass | A one-time, owner-only bypass of the ruleset, restored immediately and recorded in REX. |
| Pre-match | Reusing a result already proven for the identical commit and inputs instead of re-running it. |
| REX · KEB | CODEX evidence stores: reusable lessons and recurrence (REX), evidence identity (KEB). |
| GLOB · MESH_DOCK | Global capability census and the producer/consumer contract between repos. |
| SIPOC | Supplier, Input, Process, Output, Customer. With .self, the customer of N supplies N+1. |
| Exact head | The one commit SHA under test. Evidence for any other SHA does not count. |
| Non-waivable gate | A required check or review that must be green on the same SHA before merge. |
| Single-flight | One admitted PR merges at a time so no head goes stale behind another merge. |
| Receipt | A small record binding a result to an exact commit and input hash. |
| DOW | ABACUS derived analysis: ranks findings and returns scoped dispositions. |
| Inherited debt | A red already present on main, as opposed to a defect the PR introduced. |

THE CENTRAL RULE

# Seven steps, in order, every pulse

**01**

**Reuse proven evidence**

D→M

**02**

**Allocate, preconfigure, pretest**

M

**03**

**Find the first material red**

A

**04**

**Repair the smallest common cause**

I

**05**

**Verify independently**

I

**06**

**Publish exact-head evidence**

C

**07**

**Advance only admissible successors**

C

A passing local preflight raises confidence for the next iteration. Only exact-head GitHub checks and reviews move a PR through admission.

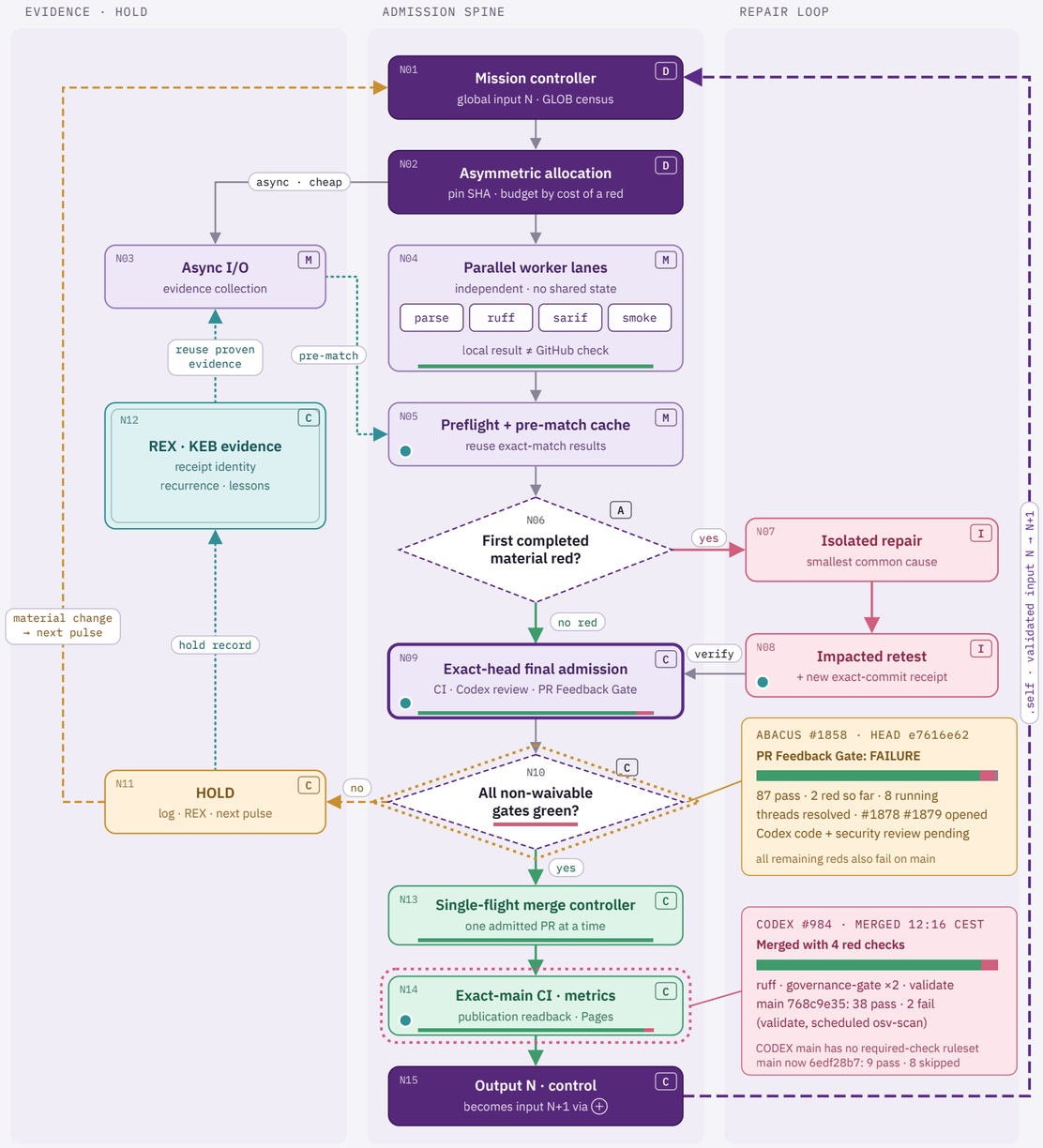
EXECUTION PULSE

# One pulse, three lanes

The admission spine runs down the centre. Reds go right into the repair loop; holds and evidence sit on the left. The dotted rings mark where the two tracked PRs are now.

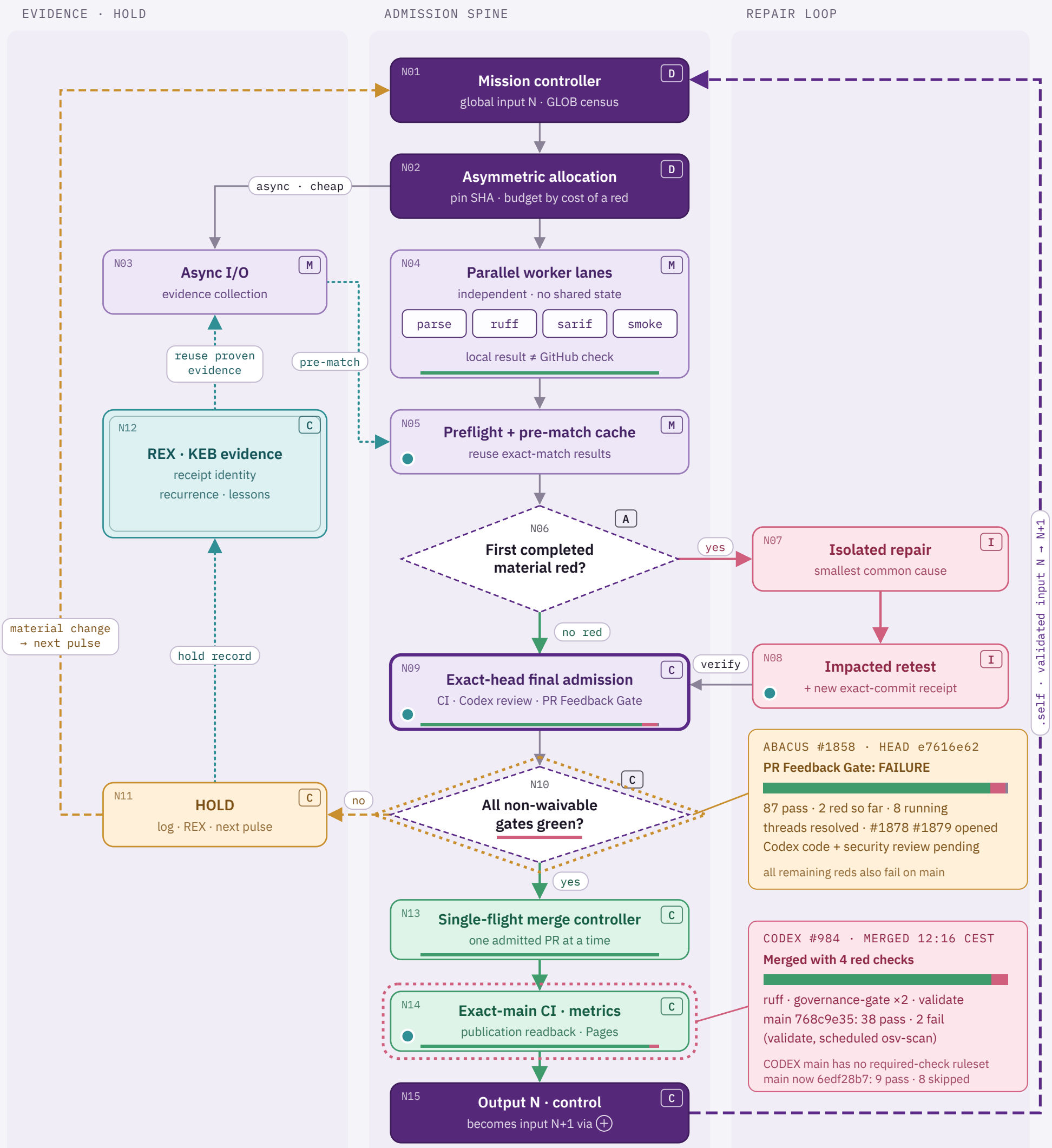
- Authority
- Gate
- Admitted path
- Evidence store
- Process
- Repair
- Hold

**#1858** waits at N10 (gate). **#984** reached N14 without passing N10.



EXECUTION PULSE · PORTRAIT DETAIL OF SLIDE 09

# One pulse, three lanes




Authority · Process · Gate · Repair
Admitted path · Hold · Evidence store · Sequence
Pass · Material red · Hold · Evidence
.self recursion

| | | | | | |
|---|---|---|---|---|---|
| N01 | Mission controller | Define | N02 | Asymmetric allocation | Define |
| N03 | Async I/O | Measure | N04 | Parallel worker lanes | Measure |
| N05 | Preflight + pre-match cache | Measure | N06 | First completed material red? | Analyse |
| N07 | Isolated repair | Improve | N08 | Impacted retest | Improve |
| N09 | Exact-head final admission | Control | N10 | All non-waivable gates green? | Control |
| N11 | HOLD | Control | N12 | REX · KEB evidence | Control |
| N13 | Single-flight merge controller | Control | N14 | Exact-main CI · metrics | Control |
| N15 | Output N · control | Control | | | |

PROCESS STAGES 1–4 · CLICK A NODE ID FOR ITS REFERENCE ROW

# From trigger to first red

## 1 DEFINE — Mobilise

N01 Mission controller N02 Asymmetric allocation

ENTRY
A trigger: new head, schedule or material change

ACTIONS
Census authorities and capabilities
Pin the exact SHA
Budget lanes by the cost of a red

EXIT
Pinned SHA pair
Lane budget

## 2 MEASURE — Collect and run

N03 Async I/O N04 Parallel worker lanes

ENTRY
SHA pinned, lanes budgeted

ACTIONS
Collect prior evidence asynchronously from REX, KEB and GitHub
Run parse, Ruff, SARIF and smoke lanes in parallel

EXIT
Evidence bundle
Per-lane results

## 3 MEASURE — Pre-match

N05 Preflight + pre-match cache

ENTRY
Lane results and evidence bundle

ACTIONS
Reuse results proven for the identical commit and inputs
Run only the remainder

EXIT
Preflight receipt (local, not CI proof)

## 4 ANALYSE — Triage

N06 First completed material red?

ENTRY
Preflight receipt

ACTIONS
Rank findings with DOW
Assign one of four failure classes
Select the first completed material red

EXIT
One red to repair, or none

PROCESS STAGES 5–8 · CLICK A NODE ID FOR ITS REFERENCE ROW

# From repair to the next pulse

## 5 IMPROVE — Repair loop

N07 Isolated repair · N08 Impacted retest

ENTRY
One classified red

ACTIONS
Fix the smallest common cause on an isolated branch
Retest the impacted set
Write a new exact-commit receipt

EXIT
Repaired head submitted for verification

## 6 CONTROL — Admission

N09 Exact-head final admission · N10 All non-waivable gates green?

ENTRY
Exact head with no known material red

ACTIONS
Run CI, Codex review and PR Feedback Gate on the exact SHA
Evaluate non-waivable gates only

EXIT
Admit, or hold

## 7 CONTROL — Hold and learn

N11 HOLD · N12 REX · KEB evidence

ENTRY
A non-waivable gate red or pending

ACTIONS
Record the reason in REX
Update recurrence and lessons
Plan the next pulse

EXIT
Hold record
Next pulse on material change

## 8 CONTROL — Merge and prove

N13 Single-flight merge controller · N14 Exact-main CI · metrics · N15 Output N · control

ENTRY
Admission green

ACTIONS
Merge one PR at a time
Run exact-main CI, record metrics, read back the publication
Write the census row

EXIT
Output N becomes input N+1

NODE REFERENCE 1 / 2 · STAGE AND TRIGGER IDS ARE LINKS

# Mobilise, measure, triage, repair

| ID | NODE · WHAT IT DOES | ROLE | STAGE | TRIGGERS | ARTEFACT | STATUS |
|---|---|---|---|---|---|---|
| N01 | **Mission controller**<br>Takes global input N, discovers authorities and capabilities, and resolves producer, consumer, receipts and scope for this pulse. | Authority | 1 | T01 T07 T12 | CODEX GLOB.yaml, .mesh/MESH_DOCK.yaml | Seeded / committed |
| N02 | **Asymmetric allocation**<br>Pins the exact head under test and spends compute where a red is most expensive. Cheap I/O runs async; costly checks get dedicated lanes. | Authority | 1 | T01 | Seed YAML execution contract | Seeded / committed |
| N03 | **Async I/O**<br>Collects existing evidence without blocking the workers, including prior receipts that may already prove a result. | Process | 2 | T02 | REX and KEB lookups, GitHub check readback | Not implemented |
| N04 | **Parallel worker lanes**<br>Runs parse, Ruff, SARIF security scans and smoke in parallel with no shared state. | Process | 2 | T02 | ABACUS scripts/testpilot_preflight.py, pre-commit, Ruff, Bandit, Semgrep | Partial |
| N05 | **Preflight + pre-match cache**<br>Merges worker output with collected evidence. A result already proven for the identical commit and inputs is reused instead of rerun. | Process | 3 | T03 | Receipts keyed by exact commit and input hash | Not implemented |
| N06 | **First completed material red?**<br>Ranks findings and selects the first completed red that matters, classified as source defect, inherited debt, review capacity or runner failure. | Gate | 4 | T03 T03b T04 | ABACUS DOW | Not implemented |
| N07 | **Isolated repair**<br>Fixes the smallest common cause only. Security-sensitive governance changes cannot approve themselves. | Repair | 5 | T05 | Smallest-scope branch | Not implemented |
| N08 | **Impacted retest**<br>Reruns only what the repair touches and writes a new exact-commit receipt. | Repair | 5 | T06 | Impacted test set | Not implemented |

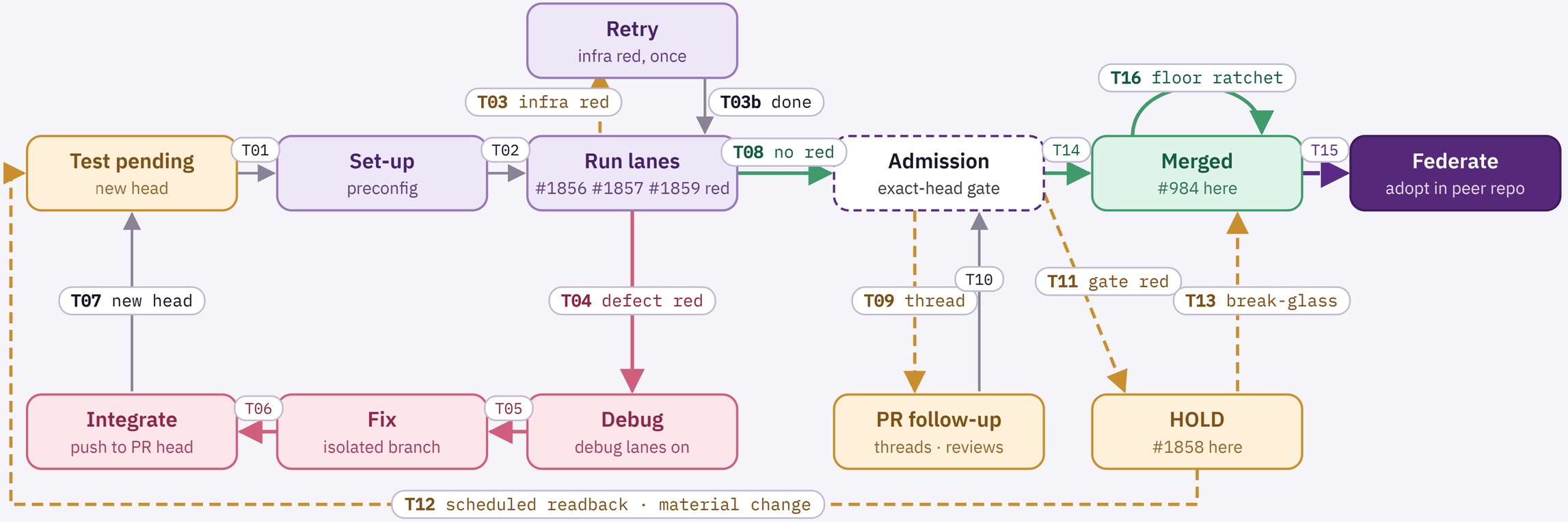
NODE REFERENCE 2 / 2 · STAGE AND TRIGGER IDS ARE LINKS

# Admit, hold, learn, merge, prove

| ID | NODE · WHAT IT DOES | ROLE | STAGE | TRIGGERS | ARTEFACT | STATUS |
|---|---|---|---|---|---|---|
| N09 | **Exact-head final admission**<br>Independent verification on the exact head SHA, including required reviews. | Process | 6 | T08 T09 | GitHub Actions, PR Feedback Gate, Codex review and security review | In progress |
| N10 | **All non-waivable gates green?**<br>Admits only when every non-waivable check and review is green on the same SHA. Waivable checks are recorded, not counted. | Gate | 6 | T10 T11 T13 | Required checks and reviews at exact head | On hold |
| N11 | **HOLD**<br>Logs the blocking reason to REX and plans the next pulse. A material change re-enters at the mission controller. | Hold | 7 | T09 T11 T12 | REX entry | On hold |
| N12 | **REX · KEB evidence**<br>Holds evidence identity, recurrence counts and reusable lessons. Async I/O reads from it at the start of each pulse. | Evidence store | 7 | T11 T16 | CODEX KEB and REX | Partial |
| N13 | **Single-flight merge controller**<br>Merges one admitted PR at a time so no head goes stale behind another merge. | Admitted path | 8 | T13 T14 | Merge queue discipline | Not reached |
| N14 | **Exact-main CI · metrics**<br>Proves main is green after the merge, records metrics and reads the published result back. | Admitted path | 8 | T15 T16 | GitHub Actions on main, Pages | Not reached |
| N15 | **Output N · control**<br>Becomes the input of iteration N+1 through governed reconciliation (⊕). | Authority | 8 | T15 | Iteration census | Not reached |

LIFECYCLE

# What moves a work item from state to state




T01 T02 T03 T03b T04 T05 T06 T07 T08 T09 T10 T11 T12 T13 T14 T15 T16

Each arrow is a trigger T01–T16: one event, one guard, one action. Tracked PRs: #1858 sits in HOLD (via T11), #984 in Merged (via T14 without a gate).

TRIGGERS 1 / 2

# Set-up, run, debug, fix, integrate

| ID | FROM → TO | TRIGGER EVENT | ACTION | NODES | WIRED TODAY |
|---|---|---|---|---|---|
| T01 | Test pending → Set-up | Pull request opened or synchronised, or a push | Start CI; mark the gate pending | N01 N02 | Partly wired<br>GitHub starts CI on these events. ABACUS's gate re-evaluates on workflow_run and review events, not on synchronize (#1858 thread, line 599). |
| T02 | Set-up → Run lanes | Preconfig receipt written | Start the parallel lanes | N03 N04 | Proposed<br>Needs the .self harness. |
| T03 | Run lanes → Retry | A lane goes red with class runner or infrastructure (5xx, runner lost, cancelled) | Re-run the failed jobs once; log a REX entry | N05 N06 | Manual<br>Re-run all jobs button on GitHub. |
| T03b | Retry → Run lanes | Retry finished | Results re-enter triage | N06 | Manual<br>Same re-run. |
| T04 | Run lanes → Debug | Red classified as a source defect that doesn't reproduce locally | Re-run that lane with debug on (debugpy, LLDB) and keep the trace | N06 | Partly wired<br>CODEX main already runs LLDB, lldb-dap and debugpy lanes (green); nothing triggers them from a red yet. |
| T05 | Debug → Fix | Root cause recorded | Branch from the exact head | N07 | Manual<br>— |
| T06 | Fix → Integrate | Impacted retest green, with an exact-commit receipt | Push to the PR head | N08 | Manual<br>— |
| T07 | Integrate → Test pending | New head on the PR | Old receipts retired; the cycle restarts at T01 | N01 | Wired<br>Push event. |
| T08 | Run lanes → Admission | All lanes terminal | Exact-head CI plus Codex code and security review | N09 | Partly wired<br>CI is automatic. The Codex review request is manual, and the allowance is exhausted. |

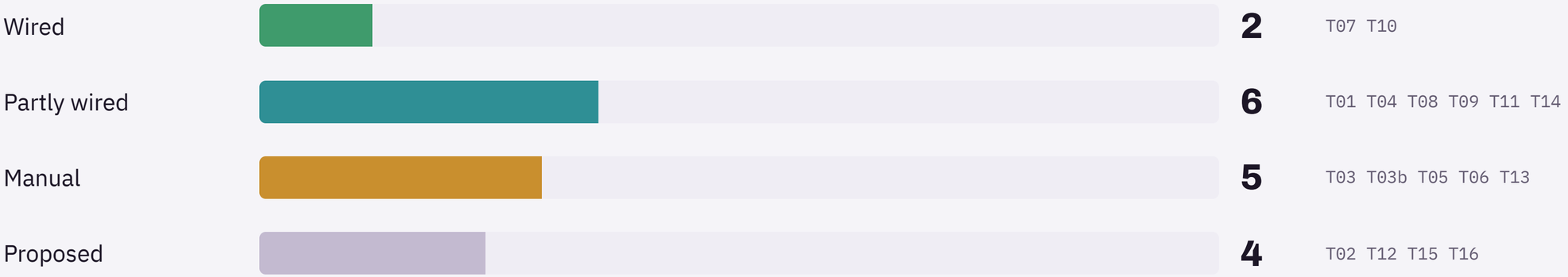
TRIGGERS 2 / 2

# Admission, follow-up, hold, merge, federate

| ID | FROM → TO | TRIGGER EVENT | ACTION | NODES | WIRED TODAY |
|---|---|---|---|---|---|
| T09 | Admission → PR follow-up | Review thread, Codex comment or gate failure | Answer, fix, or record an owner residual-risk decision | N09 N11 | Partly wired<br>Detection is wired: pr-feedback-gate.yml runs on pull_request_review and review-comment events. The response is manual. |
| T10 | PR follow-up → Admission | Thread resolved or review submitted | The gate re-evaluates | N10 | Wired<br>Review events trigger the gate. |
| T11 | Admission → HOLD | A non-waivable gate is red or pending (e.g. Codex outage) | Write the HOLD record and a REX entry | N10 N11 N12 | Partly wired<br>The gate holds. The REX record is manual. |
| T12 | HOLD → Test pending | Scheduled readback, or a material change | Start a new pulse | N11 N01 | Proposed<br>Needs a scheduled task (burndown item 7). |
| T13 | HOLD → Merged | Owner decision with a written residual risk | Squash-merge, restore the ruleset, write REX | N10 N13 | Manual<br>Owner edits ruleset 24494381. This is the proposed path for #1858. |
| T14 | Admission → Merged | All non-waivable gates green on the same SHA | Squash-merge | N13 | Partly wired<br>ABACUS: ruleset enforces it. CODEX: no ruleset; #984 merged with 4 reds. |
| T15 | Merged → Federate | Exact-main CI green and read back | Open an adoption PR in the peer repo; write the census row | N14 N15 | Proposed<br>Not built. For #984, CODEX main has 2 post-merge reds to clear first. |
| T16 | Merged → Merged | Exact-main CI measures coverage at or above floor + 1 point | Raise the coverage floor to the measured value; write the census row (total, breadth, depth, per area) | N14 N12 | Proposed<br>Not built. Today the gate is a fixed 75% and fails on every PR. |

AUTOMATION COVERAGE

# Only 2 of 17 transitions fire on their own




AUTOMATIC TODAY

Fully automatic: a new head restarts the cycle (T07) and review events re-run the gate (T10). Partly automatic: CI starts on push and PR events but the gate does not reset on synchronize (T01), and thread detection is wired while the response is manual (T09).

NEXT TO WIRE

The `.self` harness (T02 set-up receipt), a scheduled readback pulse (T12), and debug-on-red using the existing CODEX LLDB and debugpy lanes (T04).

ABACUS #1858 · HEAD E7616E62

# Gate invariants: 2 met, 3 failing, 1 running

| | | |
|---|---|---|
| ✗ | **Exact-head Codex code review complete** | Pending; Codex allowance exhausted |
| ✗ | **Exact-head Codex security review complete** | Pending; Codex allowance exhausted |
| ✓ | **Zero unresolved review threads** | Both resolved 17:20 as owner residual risk → #1878, #1879 |
| ✗ | **Current-head GitHub Actions terminal and green** | Code Quality and Security Scanning red; all inherited from main (see next slide) |
| … | **Current-head admission evidence terminal and green** | Gate re-running on e7616e62 |
| ✓ | **Known deferred warnings linked to open issues** | #1736 |

The gate checks out the default branch, so #1858's own fallback cannot admit #1858. The code reviews stay unmet until Codex is back or the owner uses break-glass.

FAILURE CLASSIFICATION

# Four reds fail on main every night: inherited, not caused by PRs

| CHECK | MAIN NIGHTLY | #1858 | #1859 | #1857 | #1856 | CLASS |
|---|---|---|---|---|---|---|
| **Code Coverage** | red | running | | red | red | Inherited debt · confirmed |
| **Code Quality Checks** | red | red | red | red | red | Inherited debt · confirmed |
| **Full Red Proof Summary** | red | running | red | red | red | Inherited debt · confirmed |
| **Security Scanning** | red | red | red | red | | Inherited debt · confirmed |
| **policy-and-inventory** | | green | | | | Source defect · fixed in e7616e62 |
| **validate-scripts** | | | red | | | Unclassified |

POLICY-AND-INVENTORY ON #1858 · FIXED

`docs/ci/WORKFLOW_RATIONALIZATION.md` went stale after the one-line policy change. Regenerated and pushed as e7616e62; the check is green on the new head.

RECURRING REDS · CONFIRMED ON MAIN

ABACUS main (53688f26) fails Security Scanning, Code Quality, Code Coverage and Full Red Proof in every nightly CI/CD Test Suite run since at least 7 October. Each is an “Enforce … red roots / threshold” step: inherited debt. #1859 targets exactly this.

LANDING #1858

# Break-glass sequence, owner and Claude steps

| | | | |
|---|---|---|---|
| 1 | Claude | Done: threads resolved, #1878 #1879 opened and corrected | 17:20 · 17:55 |
| 2 | Claude | Done: regenerated rationalization doc pushed | e7616e62 · green |
| 3 | GBO | Accept the 4 inherited reds under break-glass | they fail nightly on main |
| 4 | GBO | Add yourself as bypass actor on ruleset 24494381 | one-time |
| 5 | Claude | Squash-merge #1858, read back ABACUS main CI | merge receipt |
| 6 | GBO | Remove the bypass immediately | ruleset restored |
| 7 | Claude | Write the REX entry; re-evaluate the 20 open PRs | owner attestation now available |

#1878 and #1879 turned out to be fixed in the PR already (a7f623b5, 14430046); they stay open as post-merge checks. Ruleset at 17:55: active, 0 bypass actors.

CODEX #984

# Merged, but the admission step was skipped

WHAT HAPPENED

#984 merged at 12:16 with 4 red checks on head 90e3396a: ruff, governance-gate ×2, validate. CODEX main has no required-check ruleset, so the merge went through. After the merge, 768c9e35 had 38 pass and 2 fail. **validate** (deterministic materializations) was fixed on main at f9ae4676. **osv-scan** fails on every main push, including before #984: inherited.

WHY IT MATTERS FOR CODE PURPLE

The protocol's own rule (advance only admissible successors) was not enforced on the repository that holds the protocol. ABACUS enforces a gate that is too strict to pass; CODEX enforces none.

Decision for GBO by 14 October: give CODEX main a required-check ruleset like ABACUS, or record why not.

COVERAGE · MEASURED ON #1858 HEAD 203D63AF, 1,086 TESTS

# Coverage is 37%: breadth × depth

OVERALL

**36.9%**

line + branch, as the gate measures it

LINES

**38.2%**

10,686 of 27,998 statements

BREADTH

**64.5%**

statements in files any test touches (155 of 269 files)

DEPTH

**59.2%**

lines covered inside those touched files

```
coverage = breadth × depth = 0.645 × 0.592 = 0.38 → target 0.90 × 0.84 ≈ 0.75
```

WHY IT READS ~30%

CI recorded 37% on 1 October over a wider scope (41,408 statements). 114 files have no test at all; they hold 9,952 statements, 57% of the gap.

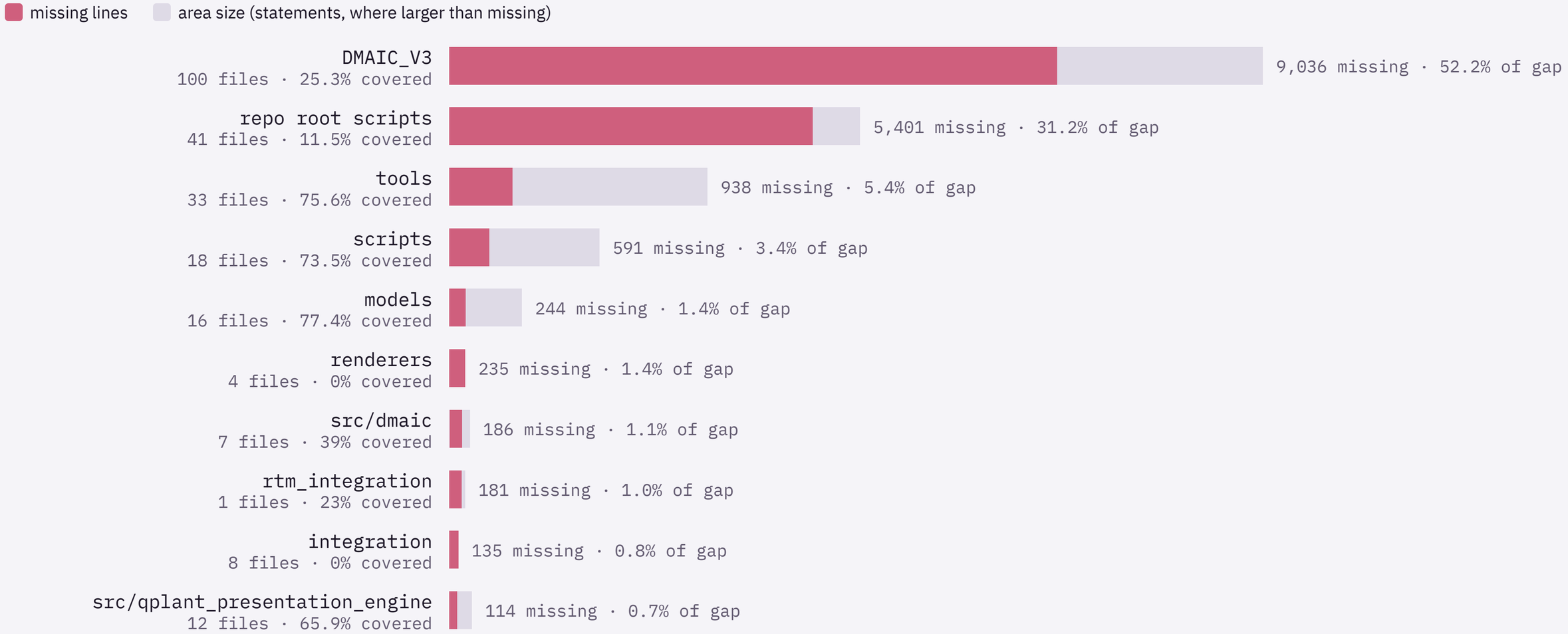
WHAT THE GATE DOES TODAY

`Enforce coverage threshold` fails at a fixed 75% on every PR, so the same red appears everywhere and steers nothing.

Local run with the CI command (`pytest --cov=.`, branch on). Docker integration tests deselected (no Docker here); CoolProp tests skip as in CI.

COVERAGE HIERARCHY · REPO → AREA → FILE

# Two areas hold 83% of the 17,312 missing lines



COVERAGE · WHAT IS DONE TODAY

# Measured on every PR, but nothing ratchets it upward

IN PLACE

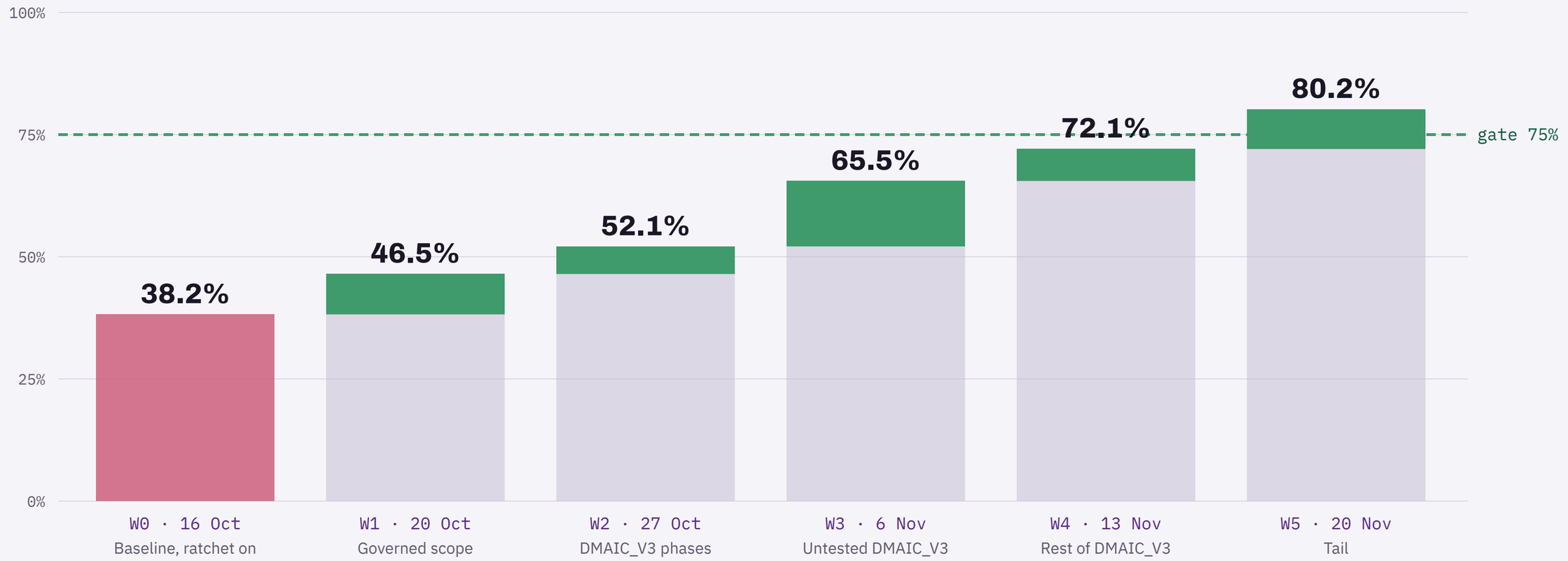
- `ci-cd-tests.yml` measures broad coverage on every PR and writes it to the job summary
- A ranked debt list per run (artifact `broad-coverage-ranking-<sha>`)
- `governance/coverage/criticality_manifest.json`: user-directed high-criticality files (B10 cluster)
- Targeted tests in `tests/coverage/` for phase 1, 4, 8, execution tracker, debug pipeline
- MIP B0 canonical coverage evidence with per-test contexts (`dynamic_context = test_function`)
- #1859 proposes a PR-level new-debt ratchet

MISSING

- No floor that rises: the gate is a fixed 75%, red on every PR
- No diff coverage on new or changed lines
- No wave plan with owners and dates
- 34 one-shot or legacy files sit in the denominator without a disposition
- Per-test contexts are recorded but not used to pick impacted tests (N05 pre-match)
- Coverage is not in the iteration census, so trends are invisible

COVERAGE · WAVE SEQUENCE (PROPOSED)

# Six waves from 38% to 80%, each one raising the floor




Line coverage, simulated from today's per-file data: each wave raises its files to the stated level. The floor only moves up after exact-main CI confirms the new value (trigger T16). W1 changes the denominator, so it needs a written disposition per file.

COVERAGE · WAVE TARGETS

# Which files each wave works on

| WAVE | FILES | SIZE | NOTE |
|---|---|---|---|
| **W1** | abacus_v21_* (17), deploy_* and run_* (6), demo, ci_monitor_local, clone_based_validator, _LEGACY_phase0_setup and others | 34 files · 5,017 statements | Each file gets keep+test, archive or omit, in writing. DOW_DEPLOYMENT_ORCHESTRATOR.py is a keep-and-test candidate |
| **W2** | phase9 · phase4 · phase1 · phase8 · phase7 · phase6 · phase2 · phase3 · phase5 · phase0_init | 1,749 missing lines | Extend tests/coverage/ phase tests; they already exist for 1, 4, 8 |
| **W3** | agents/self_improvement_agent · convergence/maturity_tracker · core/canonical_index · core/link_tracker · generators/master_reconciliation · integrations/git_manager … | 53 modules · 4,382 missing | DMAIC_V3 root 1,842 · generators 646 · core 591 · convergence 504 · agents 385 |
| **W4** | Partly covered DMAIC_V3 modules (debug_pipeline, execution_tracker …) | 34 modules | Depth work: branches and error paths |
| **W5** | tools, scripts, models, src/dmaic, src/qplant_presentation_engine, renderers, integration | 136 files | Mostly depth; renderers and integration have no tests yet |

COVERAGE IN THE PROTOCOL

# Where coverage sits in the lanes, nodes and triggers

| AREA \ WAVE | W0 | W1 | W2 | W3 | W4 | W5 |
|---|---|---|---|---|---|---|
| **DMAIC_V3** | guard | guard | work | work | work | |
| **repo root** | guard | work | | | | |
| **tools** | guard | | | | | work |
| **scripts** | guard | | | | | work |
| **models** | guard | | | | | work |
| **src/*** | guard | | | | | work |
| **renderers, integration** | guard | | | | | work |

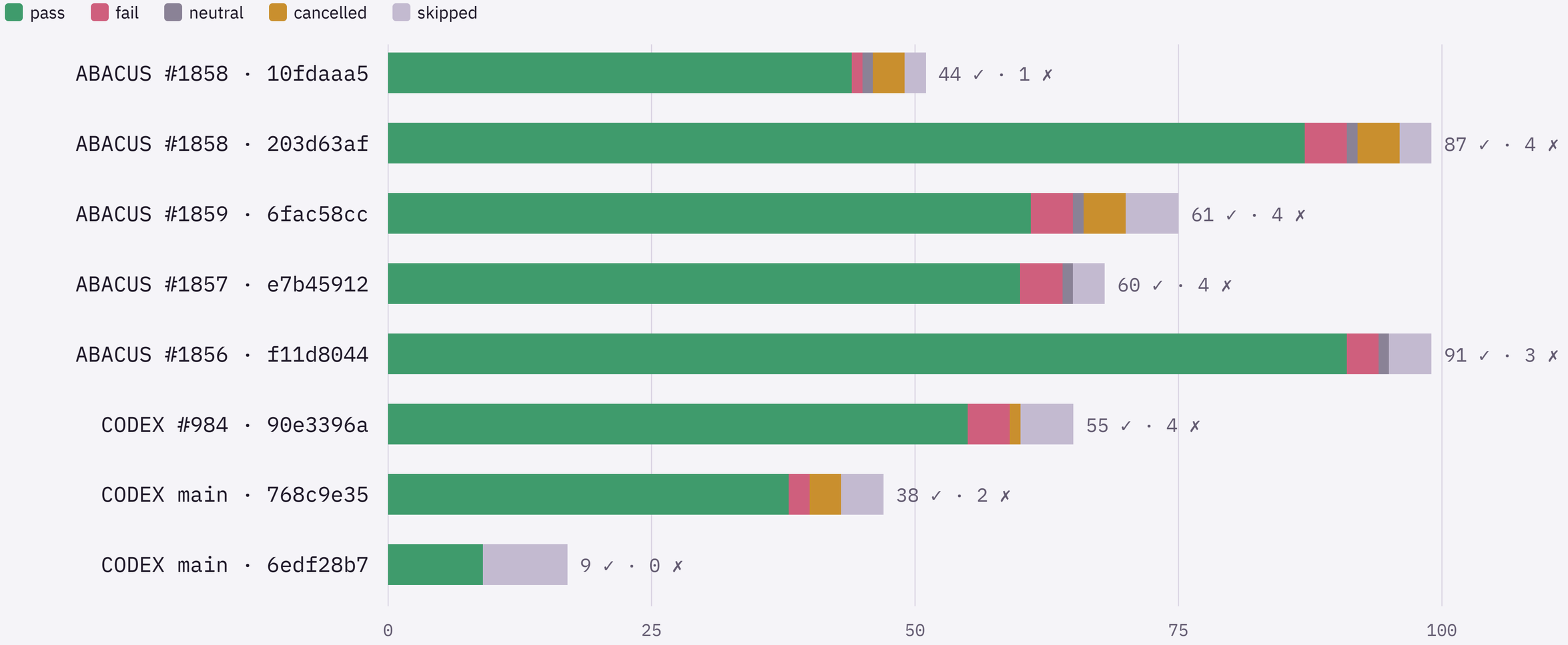
guard = ratchet and diff coverage protect the area; work = the wave adds tests or a disposition.

ATTACHMENT POINTS

| | |
|---|---|
| LANE | New **Coverage** lane in tier Verify, owned by ABACUS `ci-cd-tests` and MIP B0 |
| N04 | Worker lane measures line + branch per test |
| N05 | Pre-match uses per-test contexts to run only impacted tests |
| N06 | Fixed-threshold red = inherited debt; diff coverage below 80% = source defect |
| N14 | Exact-main CI records total, breadth, depth, per area |
| N12 · CENSUS | One coverage row per iteration; trend read back each pulse |
| T16 (NEW) | Main measured ≥ floor + 1 → raise the floor. Guard: exact-main CI green |

CENSUS

# Check outcomes per head, to scale



RUNTIME · VERSION CONTROL · DEBUG

# What ran, where, on which commit

| VERSION CONTROL | |
|---|---|
| ABACUS #1858 | fix/gate-owner-attestation-fallback @ 203d63af |
| ABACUS MAIN | 53688f26 · #1838 · 2026-10-06 |
| CODEX MAIN | 6edf28b7 · 2026-10-12 14:40 CEST |
| CODEX #984 | merge 768c9e35 · 12 commits |
| RULESET | ABACUS 24494381 · active · 0 bypass |

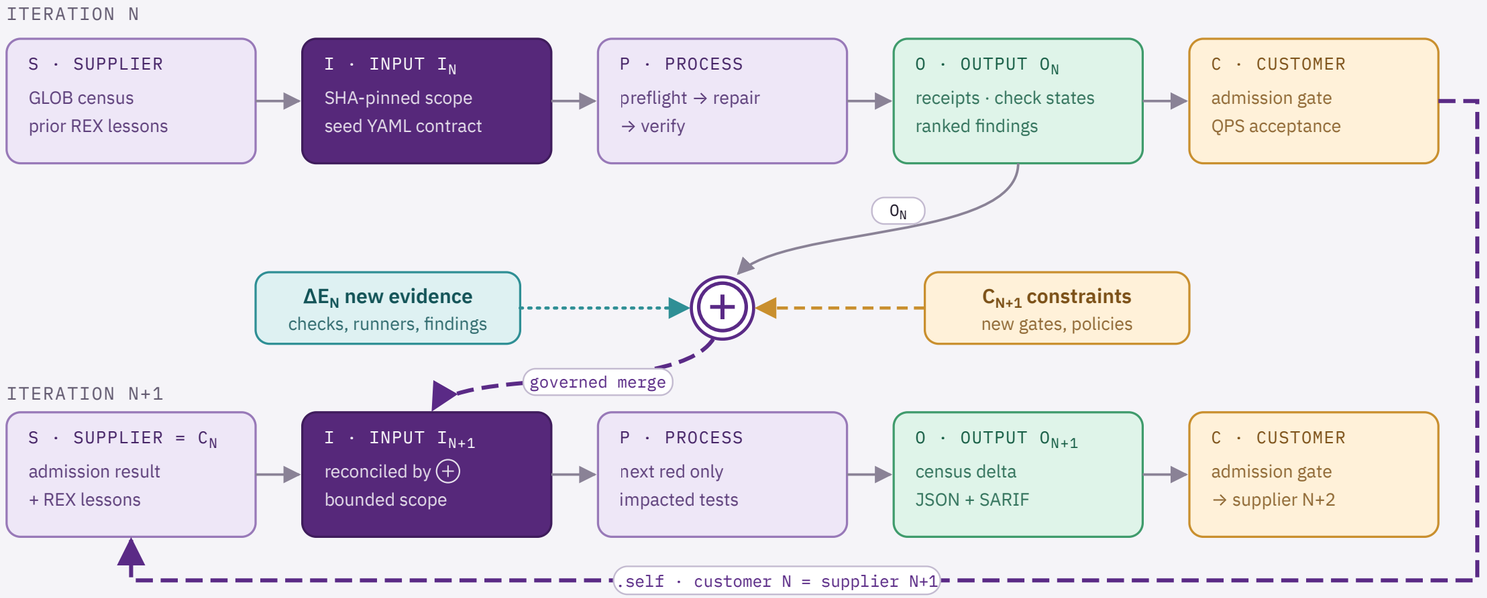
| RUNTIME AND DEBUG | |
|---|---|
| PYTHON | 3.10 · 3.11 · 3.12 (ABACUS CI matrix, all green on #1858) |
| RUNNERS | CPython (GitHub-hosted runners) · ubuntu-24.04 |
| DEBUG LANES | CODEX main: Docker real runtime LLDB, live lldb-dap + MCP, live debugpy + js-debug — all green |
| LOCAL CHECK | Python 3.13: 61 governance tests pass; inventory check passes after regeneration |
| NOT YET | .self harness census, Docker smoke, live port probing |

RECURSIVE SIPOC

# The customer of N supplies N+1

$$I_{N+1} = O_N \oplus \Delta E_N \oplus C_{N+1}$$

| | |
|---|---|
| $O_N$ | last output: receipts, check states, ranked findings |
| $\Delta E_N$ | new evidence: completed checks, runner behaviour |
| $C_{N+1}$ | new constraints: gates, policies, review rules |
| $\oplus$ | governed reconciliation, never plain concatenation |



LANES

# Nine lanes, four tiers, existing owners

**Code Purple**

| Discover | Verify | Analyse and remember | Authorise and deliver |
|---|---|---|---|
| **Global lookup**<br>GLOB.yaml · mesh census<br>CODEX | **Runtime and self-test**<br>scripts/testpilot_preflight.py<br>ABACUS | **Derived analysis**<br>DOW<br>ABACUS | **QPS domain**<br>cryoplant-project sources<br>cryoplant-project |
| **Dock interface**<br>.mesh/MESH_DOCK.yaml<br>mesh | **Lint and configuration**<br>pre-commit · Ruff · YAML/JSON/TOML<br>both | **Semantic and provenance**<br>KEB · REX<br>CODEX | **Delivery**<br>Actions · PR Feedback Gate · Pages<br>GitHub |
| | **Security**<br>Bandit · Semgrep · SARIF<br>both | | |

Runtime dock ladder: ABACUS devcontainer ports 8000 · 8080 · 5000 and the CODEX compose validator are **configured**; listening, published and probed are unproven.

BURNDOWN

# What is left, who owns it, by when

| # | ITEM | STATUS | OWNER | TARGET |
|---|---|---|---|---|
| 1 | **Land ABACUS #1858** | in progress | GBO + Claude | 13 Oct |
| 2 | **Hardening #1878, #1879** | fixed in PR | verify after merge | post-merge |
| 3 | **CODEX post-merge reds** | classified | validate fixed · OSV inherited | done |
| 4 | **CODEX admission ruleset** | decision | GBO | 14 Oct |
| 5 | **Classify recurring reds** | confirmed | 4 inherited from main | done |
| 6 | **Runnable .self harness** | todo | Claude | 16 Oct |
| 7 | **Scheduled readback pulse (T12)** | todo | Claude | 16 Oct |
| 8 | **Docker smoke, port probing** | not built | Claude | 23 Oct |
| 9 | **Cross-repo adoption (T15)** | gated on 1 | — | after #1858 |

APPENDIX

# Edge reference E01–E19

| ID | FROM → TO | TYPE | MEANING |
|---|---|---|---|
| **E01** | N01→N02 Mission controller → Asymmetric allocation | Sequence | Every pulse starts at the controller. |
| **E02** | N02→N04 Asymmetric allocation → Parallel worker lanes | Sequence | Costly checks get dedicated lanes. |
| **E03** | N02→N03 Asymmetric allocation → Async I/O | Sequence | Cheap I/O runs without blocking the workers. |
| **E04** | N04→N05 Parallel worker lanes → Preflight + pre-match cache | Sequence | Per-lane JSON and SARIF results. |
| **E05** | N03→N05 Async I/O → Preflight + pre-match cache | Evidence | Prior receipts offered for exact-match reuse. |
| **E06** | N05→N06 Preflight + pre-match cache → First completed material red? | Sequence | Merged preflight result. |
| **E07** | N06→N07 First completed material red? → Isolated repair | Material red | A completed material red exists. |
| **E08** | N07→N08 Isolated repair → Impacted retest | Material red | Repair committed on an isolated branch. |
| **E09** | N08→N09 Impacted retest → Exact-head final admission | Sequence | Retest receipt goes to exact-head verification. |
| **E10** | N06→N09 First completed material red? → Exact-head final admission | Pass | No material red among completed checks. |
| **E11** | N09→N10 Exact-head final admission → All non-waivable gates green? | Sequence | Check and review states on the exact head. |
| **E12** | N10→N11 All non-waivable gates green? → HOLD | Hold | A non-waivable gate is red or pending. |
| **E13** | N10→N13 All non-waivable gates green? → Single-flight merge controller | Pass | Every non-waivable gate is green on the same SHA. |
| **E14** | N13→N14 Single-flight merge controller → Exact-main CI · metrics | Pass | One PR merged at a time. |
| **E15** | N14→N15 Exact-main CI · metrics → Output N · control | Pass | Main green; metrics and publication read back. |
| **E16** | N11→N12 HOLD → REX · KEB evidence | Evidence | Blocking reason recorded. |
| **E17** | N12→N03 REX · KEB evidence → Async I/O | Evidence | Lessons and receipts reused in the next pulse. |
| **E18** | N11→N01 HOLD → Mission controller | Hold | A material change triggers the next pulse. |
| **E19** | N15→N01 Output N · control → Mission controller | .self recursion | Validated output becomes input N+1 through ⊕. |

SOURCES

# Where the figures come from

- ABACUS #1858 · checks, gate output, review threads (GitHub API, 17:27)
- ABACUS #1878 and #1879 · follow-up hardening issues
- #1859, #1857, #1856 · check-run tallies
- CODEX #984 · merge record and check runs; CODEX main 768c9e35 and 6edf28b7
- ABACUS ruleset 24494381 (Main Hard Gate) · `.github/workflows/pr-feedback-gate.yml`, `ci/governance/pr_feedback_gate.py`
- Code Purple execution seed · `federation/mesh/CODE_PURPLE_EXECUTION_SEED_20261012.yaml`

LIFT · 2026-10-18 13:17 UTC READBACK

# Not lifted: 2 of 5 criteria met

OPEN ABACUS PRS

**20 → 12**

baseline at declaration → now

MERGES SINCE #1858

**97**

gate green on 0; 97 with red checks

COVERAGE (LINE + BRANCH)

**42.6%**

recorded 15 Oct on main 803700a3; from 36.9%

RULESET 24494381

**incomplete**

bypass=6, rules=3; missing: required_status_checks rule absent

| ID | LIFT CRITERION | MET | EVIDENCE |
|---|---|---|---|
| L1 | **Unblocker PR merged** | ✓ | merged_at=2026-10-12T16:17:07Z |
| L2 | **Gate restored (ruleset active, no bypass, expected rules present)** | ✗ | active, bypass=6, rules=3; missing: required_status_checks rule absent |
| L5 | **Admission integrity (no merge since the unblocker without a green gate)** | ✗ | 97 merged; gate green on 0; 97 merged with red checks |
| L3 | **Merge flow restored (at least 5 merges to the default branch since the unblocker)** | ✓ | 97 merges since 2026-10-12T16:17:07Z |
| L4 | **Open PR backlog at most a quarter of the baseline** | ✗ | 12 open (baseline 20) |

The 13 Oct check passed L2 falsely: it read enforcement and bypass actors but not rule content. Fixed (EVR-ABACUS-0001). Follow-ups #1878 and #1879 verified on main. Red nightly on main: Code Quality Checks, Full Red Proof Summary, Security Scanning, Unit Tests (Python 3.10, macos-latest), Unit Tests (Python 3.10, ubuntu-latest), Unit Tests (Python 3.10, windows-latest), Unit Tests (Python 3.11, macos-latest), Unit Tests (Python 3.11, ubuntu-latest), Unit Tests (Python 3.11, windows-latest), Unit Tests (Python 3.12, macos-latest), Unit Tests (Python 3.12, ubuntu-latest), Unit Tests (Python 3.12, windows-latest).

AFTER-ACTION

# What worked, what to change next time

WORKED

- **Readback before action.** It showed #1858 could not admit itself (the gate runs main's code) and that #984 had merged red.
- **Classify against main.** Four reds were red in main's nightly run, so accepting them under break-glass rested on evidence.
- **Smallest fix.** A 3-line regenerated doc cleared the only red the PR caused.
- **Owner / agent split.** Owner: bypass, merge, restore. Agent: threads, issues, fix, readback.
- **Measurable lift.** Criteria read from GitHub; one of them was too weak (below).

CHANGE

- **Break-glass deleted the rules instead of adding a bypass, and the lift check did not read rule content.** 61 PRs merged ungated. Use a bypass actor; L2 now compares rules; L5 audits every merge.
- **Snapshots went stale twice in an hour.** Run the readback every pulse and on a schedule.
- **A decision was written against an old head.** Threads were resolved as "residual risk" minutes after fixes were pushed. Re-read the head before writing any decision.
- **The seed repo has no gate.** CODEX #984 merged with 4 reds. Decide on a CODEX ruleset.
- **One reviewer was a single point of failure.** The owner-attested fallback is now on main; test it on the next outage.
- **Inherited reds stay red.** Coverage W0 floor and diff coverage due 16 Oct.

REUSABLE METHOD

# Code Purple in eight steps

| STEP | NAME | ENTRY | ACTION | WHO |
|---|---|---|---|---|
| 0 | **Declare** | Merge flow stalled, the same checks red across PRs, or a required reviewer unavailable | Write incident.yaml: repos, gate, tracked PRs, owner, lift criteria | Owner |
| 1 | **Read back** | incident.yaml exists | Run readback.py; no action before the first census | Agent |
| 2 | **Classify every red** | Census on exact heads | Inherited (also red on main nightly) · source defect · review capacity · runner or infra | Agent |
| 3 | **Find the unblocker** | Classes known | Smallest change that lets the gate admit again; prove it cannot admit itself | Agent + owner |
| 4 | **Repair source defects only** | Unblocker chosen | Fix on the unblocker branch, impacted retest, exact-head readback | Agent |
| 5 | **Admit or break glass** | Only inherited or capacity reds left | Owner writes residual risk, adds a one-time bypass actor (never deletes rules), merges, removes the bypass, re-reads rule content | Owner |
| 6 | **Lift check** | Unblocker merged | readback.py evaluates L-criteria; census row saved | Agent |
| 7 | **Close out** | Lifted | Verify follow-ups on main and close; REX lesson; inherited reds become dated burn-down items | Agent + owner |

Rule that never bends: the owner alone breaks glass, once, in writing, and restores the gate immediately. Nothing that changes a gate approves itself.

ADAPTING IT

# One file to adapt; the same steps elsewhere

INCIDENT.YAML KNOBS

| | |
|---|---|
| `repos[].ruleset_id, gate_check` | Which gate is blocked and which single check it requires |
| `repos[].nightly_workflow` | The scheduled run on the default branch that defines the inherited set |
| `tracked_prs[].role` | unblocker · debt-ratchet · queued · seed: decides lift and report order |
| `lift_criteria[]` | pr_merged · ruleset_clean · merges_since · open_prs_at_most, with thresholds |
| `follow_up_issues[].check` | What to verify on main before closing each follow-up |
| `baseline` | Numbers at declaration, for the before/after census |

| CONCEPT | CODE (THIS CASE) | ENGINEERING DOCUMENTS | DATA PIPELINE |
|---|---|---|---|
| **Gate** | Required CI check | Approval signature on a controlled document | Validation step before a dataset is published |
| **Inherited red** | Red on main nightly | Finding already open in the register | Check already failing on the last good release |
| **Unblocker** | PR that restores admission | Revision that restores reviewability | Fix that lets validation run again |
| **Break-glass** | One-time ruleset bypass | Documented concession, owner-signed | Manual release with sign-off |
| **Lift** | Merges flow, backlog down | Reviews close, queue down | Releases resume |

Only the code column has a working readback today; the other two are mappings for reuse.

TOOLKIT

# What runs the method

| FILE | DOES |
|---|---|
| **kit/incident.yaml** | The instance: repos, gate, tracked PRs, owner, baseline, lift criteria. The only file to edit. |
| **kit/readback.py** | Reads GitHub through `gh`; classifies reds against the main nightly run; checks lift criteria; writes `census.json` and `after_action.md`. |
| **index.html** | The interactive page: pulse graph, lifecycle triggers, census, admission board. |
| **build_deck.js · merge_deck.py** | This 16:9 deck and its bookmarks, from the same model. |
| **code-purple.md** | Markdown parent with the same anchors as the PDF links. |

Run: `python kit/readback.py kit/incident.yaml` → `kit/out/census.json`, `kit/out/after_action.md`. Lifted when every L-criterion is met.